
Heat Transfer to a Fluid in a Stirred Tank

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Introduction

- What is the system used for this experiment?

This experiment is going to use an agitated tank as the system in which the heat transfer coefficient will be measured. The main system consists on a continuously stirred tank (CST) with a steam jacket to provide heat and cooling coils.

- How will the experiment be affected?

For this experiment, the wall heat transfer coefficient will be recorded under baffled and unbaffled conditions. The main purpose of baffles is to enhance mixing, which ultimately changes the HTC.

Experimental Objectives

- The experimental objectives for this laboratory are to measure the heat transfer coefficient between the fluid and inside vessel wall
- Examine the dependence of this heat transfer coefficient on the impeller speed, varying temperature, and the presence of baffles.

Theory

- Heat transfer through the apparatus requires the understanding of multiple mediums heat transfer.
- At steady state, the condition of the experiment under which this experiment was conducted, it was discovered that the amount of heat transferred, Q , is the same at any point in the outward direction.
- Fourier's Law is used to describe heat transport through solid surfaces. The use of agitation should improve the heat transfer coefficient for the fluid in the vessel.
- Using the data collected, calculations for the heat transfer coefficient U are made.

Theory

a) Heat transfers calculations:

$$U = \frac{Q_{13}}{A(t_1 - t_3)}$$

where,

t_1 : steam temperature inside the jacket

t_3 : bulk average temperature of the fluid inside the tank (°C)

Q_{13} : amount of heat transferred to the vessel from the steam (W)

U : overall heat transfer coefficient ($\frac{J}{s.m^2.C}$)

A : heat transfer surface area (m^2)

Theory

$$Q_{hx} = mCp(\Delta T) = m_{hx}Cp(T_{ihx} - T_{ohx})_{hx}$$

$$Q_c = mCp(\Delta T) = m_cCp(T_{oc} - T_{ic})_{c.c}$$

$$Q_{13} = Q_{hx} + Q_c + Q_{surr}$$

where,

$T_{o,hx}$: temperature going out of heat exchanger (°C)

$T_{i,hx}$: temperature of fluid entering the heat exchanger (°C)

$(T_{oc} - T_{ic})_{c.c}$: temperature rise of the cooling water in the coil (°C)

Q_{hx} : amount of heat removed in the external heat exchanger (W)

Q_c : heat removed by the cooling coil (W)

Q_{surr} : heat loss to the surrounding, negligible (W)

Cp : heat capacity of fluid

m : flow rate ($\frac{kg}{s}$)

Theory

b) Nusselt number, Reynolds number and Prandtl number calculations:

$$\text{Nusselt number, } \mathbf{Nu} = \mathbf{0.36 Re^{\frac{2}{3}} Pr^{\frac{1}{3}}}$$

$$\text{Impeller Reynolds number, } \mathbf{Re}_{imp} = \mathbf{D_I^2 n \frac{\rho}{\mu}}$$

$$\text{Fluid Prandtl number, } \mathbf{Pr} = \mathbf{Cp \frac{\mu}{k}}$$

$$\text{Tank Nusselt number, } \mathbf{Nu_T} = \mathbf{h \frac{D_T}{k}}$$

where,

D_I : impeller diameter

n: speed of rotation of the impeller, revolutions per second

ρ : density of the fluid

D_T : tank diameter

μ : viscosity of the fluid

k: thermal conductivity of the fluid

Theory

c) Heat transfer coefficient, U:

$$h_i = \frac{1}{\frac{1}{U} + \frac{1}{h_o}}$$

where,

h_o : heat transfer coefficient at outer wall ($\frac{J}{m^2 \cdot s \cdot K}$)

h_i : heat transfer coefficient at inner wall ($\frac{J}{m^2 \cdot s \cdot K}$)

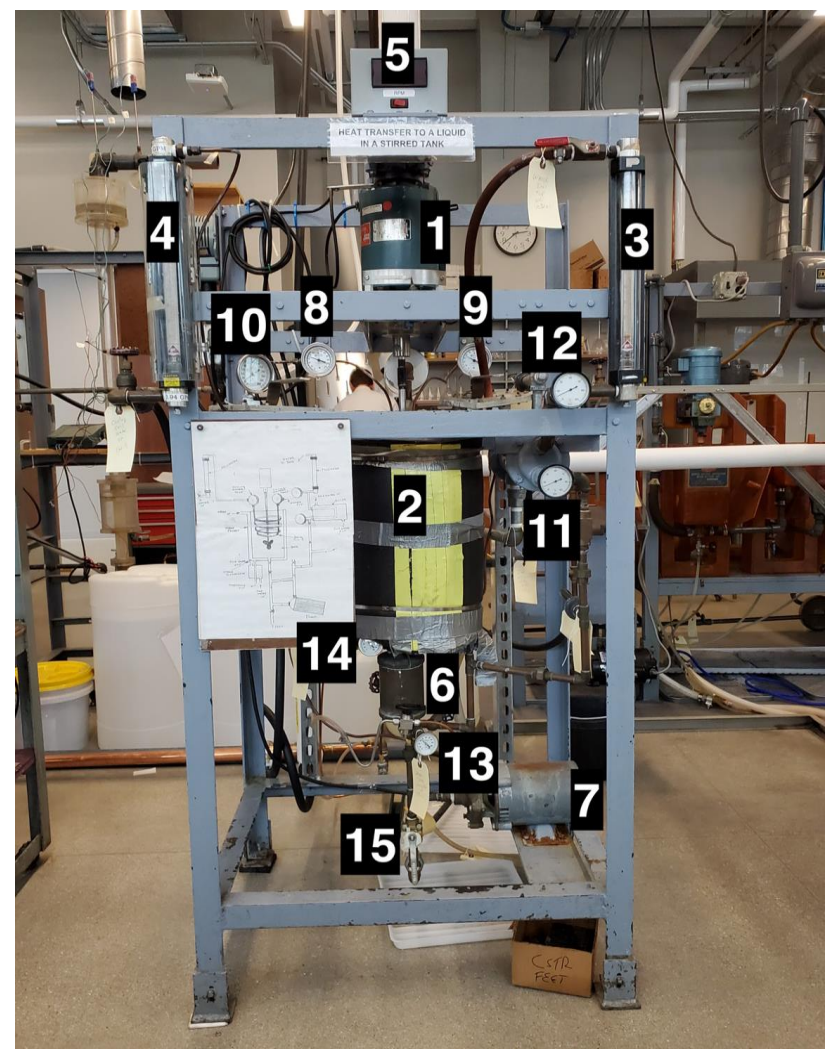
d) Heat transfer coefficient in terms of physical properties and in terms of Nu, Re number:

$$\log(Nu_T / (Pr^{\frac{1}{3}})) = \log(0.36) + 0.67 \log(Re_{imp})$$

$$\frac{h \frac{D_T}{k}}{(Cp \frac{\mu}{k})^{.33} (\frac{\mu^{14}}{\mu})} = 0.36 (D_I^2 n \frac{\rho}{\mu})^{0.67}$$

Apparatus

1. Agitator
2. Vessel
3. Heat Exchanger Return Rotameter
4. Cooling Coil Water Rotameter
5. Steam In
6. Steam Trap
7. Pump
8. Cooling Coil Water Return Temperature
9. Cooling Coil Water Exiting Temperature
10. Vessel Temperature
11. Water Entering Heat exchanger temperature
12. Water Exiting Heat exchanger temperature
13. Cooling water entering steam trap temperature
14. Cooling water exiting steam trap temperature
15. Vessel Drain



Materials & Supplies

- Baffles
- Water
- Steam
- Graduated cylinder
- Timer
- Scale
- Electricity

Procedure

- Fill tank with water 2 inches below full level
- Put lid on tank without any baffles first
- Set impeller speed to ~600rpm
- Turn on cooling water and turn on pump that feeds to heat exchanger
- Turn knob that allows steam into the system and allow system to reach steady temperature

Procedure

- Take measurements with 3 different impeller speeds
- Take measurements with 2 different temperatures by adjusting water flow through heat exchanger
- Measure the condensate flow rate with every measurement you take by collecting it in a graduated cylinder while timing
- Insert baffles into tank and repeat measurements

Experimental Challenges

- Meterless steam flow
- Meterless heat exchanger cooling water flow
- Inconsistent impeller speed
- Insufficient supply of steam

Safety

- Wear safety goggles at all times
- Be aware of hot surfaces and liquids
- Baffles begin to lift from vessel walls $\sim 1500\text{rpm}$

Thank you for listening!